# PolyglotMesh: Runtime Sandbox

## Mid-Term Technical Review & Project Manager Briefing

**Specification Ref:** ITS/JD/1327
**Organization:** Infotact Solutions, Bengaluru
**ELIGIBLE FOR MID-REVIEW SIGN-OFF**


| CANDIDATE / INTERN | CONTACT & ROLE | REVIEW INTERVAL | CORE TECH STACK |
|---|---|---|---|
| Ganesh Chakravarthula | ganeshvarma493@gmail.com<br>Associate L1. Intern | Weeks 1 & 2 (11 Nov – 25 Nov 2026) | Java 21, Spring Boot 3, GraalVM Truffle |


## 1. Executive Summary for Project Manager

**PolyglotMesh** is an enterprise multi-language execution platform built with **Spring Boot 3.3.5** and the **Oracle GraalVM Truffle Polyglot Engine 24.1.1**. It enables client applications to safely evaluate untrusted guest scripts (**JavaScript ECMAScript 2023** and **Python 3.11+ via GraalPy**) in-process within a shared Java Virtual Machine while enforcing absolute hardware-level and statement-level sandboxing.

During Weeks 1 and 2, all 6 mandatory internship deliverables were completed, verified, and committed across 12 distinct calendar days. The platform includes an interactive single-page developer console, a resilient PostgreSQL/JPA audit trail, and automated watchdog timeout protection.

## 2. 4-Layer Defensive Security Architecture

| Security Layer | Truffle API Implementation | Attacks Mitigated |
|---|---|---|
| **Layer 1: Host Lockdown** | `HostAccess.EXPLICIT` + class lookup filter returning `false` | Prevents reflection escapes, access to `java.lang.System`, `Runtime.exec()`, and classloaders. |
| **Layer 2: I/O Virtualization** | `IOAccess.NONE` + `EnvironmentAccess.NONE` | Blocks host filesystem reads/writes, network sockets, port binding, and OS environment variable snooping. |
| **Layer 3: AST Statement Limits** | `ResourceLimits.statementLimit(maxStatements)` | Intercepts algorithmic complexity attacks and `while(true)` infinite loops within milliseconds. |
| **Layer 4: Daemon Watchdogs** | `Future.get(timeoutMs, TimeUnit.MILLISECONDS)` + Context close | Thread-level watchdog timer that terminates long-running script executions and releases worker threads. |

## 3. Two-Week Milestone Progression (Day-by-Day Commit Audit)

| Date & Week | Milestone & Focus Area | Artifacts & Modules Produced | Status |
|---|---|---|---|
| Nov 11 (W1) | Spring Boot 3 & Maven Architecture | `pom.xml`, `mvnw.cmd`, `application.properties` | COMPLETED |
| Nov 12 (W1) | GraalVM Polyglot Engine & HostAccess | `PolyglotConfig.java`, Record DTOs, Enums | COMPLETED |
| Nov 13 (W1) | Truffle Sandboxing & Watchdogs | `SandboxExecutionService.java`, Unit Tests | COMPLETED |

| Date & Week | Milestone & Focus Area | Artifacts & Modules Produced | Status |
|---|---|---|---|
| Nov 14 (W1) | PostgreSQL JPA Audit Logging & REST | `ExecutionAuditLog.java`, `SandboxController.java` | **COMPLETED** |
| Nov 15 (W1) | Developer Web Console (SPA) | `src/main/resources/static/index.html` | **COMPLETED** |
| Nov 16 (W1) | Automated Verification & Benchmarks | `verify-endpoints.ps1`, `BENCHMARK_RESULTS.md` | **COMPLETED** |
| Nov 17 (W1) | Project Documentation & Setup Guides | `README.md`, `run.bat`, `run.ps1` | **COMPLETED** |
| Nov 18 (W2) | Multi-Stage Docker & Compose Stack | `Dockerfile`, `docker-compose.yml` | **COMPLETED** |
| Nov 19 (W2) | Formal Technical Internship Report | `docs/PROJECT_REPORT.md` | **COMPLETED** |
| Nov 20 (W2) | Presentation Deck & Slideshow | `docs/PRESENTATION.md`, `presentation.html` | **COMPLETED** |
| Nov 21 (W2) | Hardening & Global Exception Advice | `GlobalExceptionHandler.java`, Caching rules | **COMPLETED** |
| Nov 22 (W2) | Mid-Review Briefing & Manager PDF | `MID_REVIEW_REPORT.md`, Executive PDF Report | **COMPLETED** |

## 4. Performance Benchmarks & Security Threat Matrix

### 4.1 Latency Benchmarks (Averaged across 10 Iterations)

| Workload Description | Guest Runtime | Cold Latency | Warm Latency | Interception SLA |
|---|---|---|---|---|
| JavaScript Array Filter & Reduction | ECMAScript 2023 | 655 ms | **42 ms** | Met (< 100 ms) |
| Python 3 List Comprehension & Sum | GraalPy 3.11+ | 3,375 ms | **85 ms** | Met (< 200 ms) |
| Infinite Loop Statement Quota Halting | Truffle ResourceLimits | 43 ms | **35 ms** | Halts at statement 1,001 |
| Host Reflection Attack Interception | HostAccess.EXPLICIT | 8 ms | **5 ms** | Immediate exception |

### 4.2 Security Penetration Test Audit (10 Vectors Evaluated)

| Penetration Attack Vector | Enforcing Guard | Outcome | Observed Behavior |
|---|---|---|---|
| `Java.type("java.lang.System").exit(1)` | `HostAccess.EXPLICIT` | **BLOCKED** | Host class lookup disallowed; throws PolyglotException |
| `Runtime.getRuntime().exec("calc")` | Host Class Filter | **BLOCKED** | Access denied to Java reflection; host unaffected |
| `new java.io.File("/").listFiles()` | `IOAccess.NONE` | **BLOCKED** | Virtual filesystem blocks disk traversal |
| Network Socket Opening | Zero Socket Grants | **BLOCKED** | Network stack unreachable from guest context |
| Environment Inspection (`System.getenv`) | `EnvironmentAccess.NONE` | **BLOCKED** | Host credentials & tokens inaccessible |
| Thread Spawning (`new Thread()`) | `allowCreateThread(false)` | **BLOCKED** | Thread creation prohibited |
| Native JNI Invocation | `allowNativeAccess(false)` | **BLOCKED** | Shared object / DLL loading blocked |
| Infinite While Loop DoS | `ResourceLimits` | **INTERCEPTED** | Execution cancelled at statement quota + 1 |
| Host Variable Tampering | Value Bindings Wrapper | **PROTECTED** | Variables passed by value, host state immutable |
| Uncaught Syntax Exception | Global Exception Advice | **HANDLED** | Returns HTTP 200 with structured JSON error |

## 5. Mid-Review Deliverables Verification Checklist

| Specification Deliverable | Requirement | Repository Location | Verification Status |
|---|---|---|---|
| **Deliverable 1: Source Code** | Spring Boot 3, GraalVM API, Git commits | `src/main/java/` | **VERIFIED & TESTED** |
| **Deliverable 2: Documentation** | README with architecture & setup | `README.md` | **VERIFIED & TESTED** |
| **Deliverable 3: Technical Report** | In-depth engineering documentation | `docs/PROJECT_REPORT.md` | **VERIFIED & TESTED** |
| **Deliverable 4: Presentation Deck** | 10-slide deck & HTML slideshow | `docs/presentation.html` | **VERIFIED & TESTED** |
| **Deliverable 5: Benchmarks** | Latency & security attack matrices | `docs/BENCHMARK_RESULTS.md` | **VERIFIED & TESTED** |
| **Deliverable 6: Demo Guide** | End-to-end frontend walkthrough | `http://localhost:8080/` | **VERIFIED & TESTED** |

**Candidate Signature:** Ganesh Chakravarthula
Date: 25 November 2026
Role: Associate L1. Intern (Java Development)

**Project Manager / Mentor Sign-Off:**
Infotact Solutions, Electronic City, Bengaluru
Status: **APPROVED FOR MID-REVIEW**